

Supplementary Information

Phase-based resolution of ill-conditioned singularity ladders in nonlinear fluid models

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Every number in this document is generated from the logged outputs in `data/outputs` of the code and data package: the tables are written by `code/analysis/make_si_tables.py` and the figures by `code/figures/make_figures.py`. Evidence levels: [P] proved (elementary), [F] formal matched asymptotics, [N] numerical with stated error.

Supplementary Note 1. Validation on 2D Boussinesq and the Hou–Luo model

The pipeline used for IPM was first developed and validated on the 2D Boussinesq system with a boundary, the standard proxy for axisymmetric Euler at a wall[1], and on its one-dimensional boundary restriction, the Hou–Luo model[2]. In both, $\varepsilon = \lambda - 1$ in equation (1) of the main text and the branch is followed in $z = 1/(\lambda - 1)$.

Profiles and two independent solvers (2D) [N]. The marching solver finds eight smooth profiles $\lambda_0, \dots, \lambda_7$ (Supplementary Table 1). The independent global solver (Methods), which shares no numerical ingredient with it, reproduces all eight; with matched origin truncation the differences are at most 4.6×10^{-6} . The deepest rungs are weakly determined ($|m - 2| \lesssim 10^{-6}$ nearby), the same exponential wall as in IPM.

Supplementary Table 1. 2D Boussinesq profiles. Marching solver ($h_s = 0.0125$ from $n = 4$); global solver extrapolated in $s_{\max}$ (difference global – marching). Source: `programs/P03_boussinesq_ladder` (`glob_*`, `bq_*` logs).

n	λ_n	$z_n = 1/(\lambda_n - 1)$	$z_n - z_{n-1}$	global – marching
0	1.9205593	1.08630	—	$+1.7 \times 10^{-6}$
1	1.3990961	2.50566	1.41937	-1.1×10^{-7}
2	1.2523487	3.96277	1.45711	-5.6×10^{-7}
3	1.1842533	5.42733	1.46455	-3.2×10^{-7}
4	1.1449857	6.89723	1.46991	-4.2×10^{-7}
5	1.1194738	8.37004	1.47281	$+1.1 \times 10^{-7}$
6	1.1015817	9.84429	1.47426	-4.6×10^{-6} to $+0.4 \times 10^{-6}$
7	1.0883384	11.32010	1.47581	$+2.1 \times 10^{-7}$ ($s_{\min} = -20$)

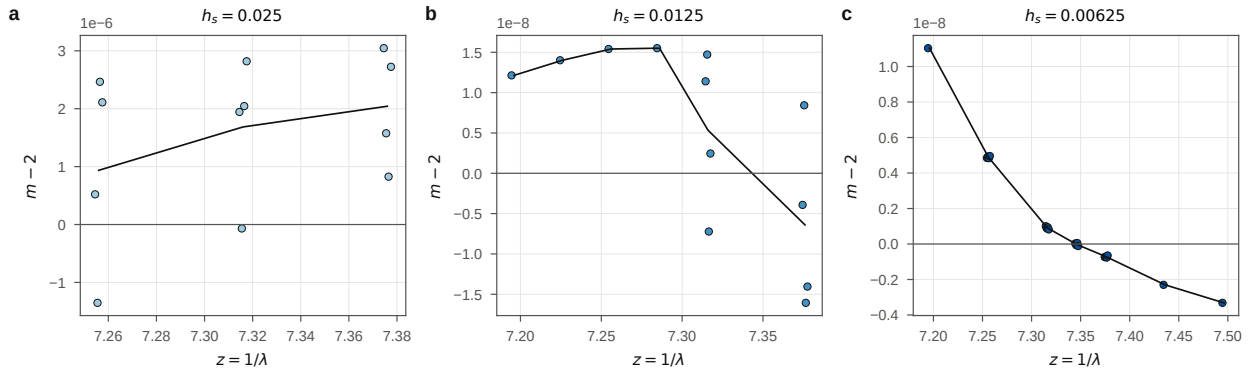
Instability index [N]. Perturbations grow as $e^{\mu\tau}$, $\tau = -\ln(1 - t)$. Method 1 linearizes the march and Biot–Savart map; μ is an eigenvalue if the linearized map T_μ has eigenvalue 1. Right-half-plane counts by the argument principle applied to $\det(I - T_\mu)$, which include the trivial time-translation mode $\mu = 1$, give 1.997, 3.000, 4.000, 5.000, 5.952, 6.942 and 7.921 zeros for the 2D profiles $n = 1$ –7: the n -th profile has exactly n

unstable directions, all real, and the counts are unchanged under domain and grid variants. Method 2, a shift-invert eigensolver on the global discretization, reproduces 27 of the 28 unstable eigenvalues of $\lambda_1\text{--}\lambda_7$ to 5×10^{-5} ; the remaining one lies in its truncation-artefact band and is confirmed by method 1 with two truncations. In the Hou–Luo model (11 profiles) the index is n for $n \leq 10$. In IPM, method 1 gives contour counts 0.994, 1.947, 3.082, 4.077 and 4.902 for $n = 0\text{--}4$ (`ipm_contour_U*.out`), i.e. index n .

Spacings [N]. For a stalled layer of fixed extent (2D Boussinesq, Hou–Luo) the spacings of $1/\varepsilon_n$ increase towards a constant; for IPM, whose layer recedes, they contract (Supplementary Fig. 3).

Supplementary Note 2. Error audit

Front-grid locking. Supplementary Table 2 lists $m - 2$ on radial grids shifted by δh_s around the seventh profile, and Supplementary Fig. 1 shows them. The scatter over shifts is the locking error at that grid; its half-range falls from $\sim 10^{-6}$ ($h_s = 0.025$) to $\sim 10^{-8}$ (0.0125) and $\sim 10^{-10}$ (0.00625), i.e. as $h_s^{7.1}$ (main-text Fig. 3a). On $h_s = 0.025$ the scatter exceeds the defect itself, which is why that grid produced a spurious zero near $z = 6.905$ (pre-registration record, stage 2d).



Supplementary Fig. 1. Raw grid-shift data near the seventh profile. $m - 2$ on four radial grids shifted by δh_s ($\delta = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$; symbols offset slightly in z by δ) and their mean (line), for $h_s = 0.025$ (a), 0.0125 (b) and 0.00625 (c). Only on the finest grid does the scatter fall below the defect.

Supplementary Table 2. Grid-shift data (`rich7_*.out`; unshifted values from the scans `ipm_scan_{p7,h7}`).

z	h_s	$\delta = 0$	$\delta = \frac{1}{4}$	$\delta = \frac{1}{2}$	$\delta = \frac{3}{4}$	half-range
7.256	0.00625	4.85×10^{-9}	4.93×10^{-9}	4.84×10^{-9}	4.96×10^{-9}	6.0×10^{-11}
7.256	0.025	5.21×10^{-7}	-1.35×10^{-6}	2.46×10^{-6}	2.11×10^{-6}	1.9×10^{-6}
7.316	0.00625	1.00×10^{-9}	8.80×10^{-10}	9.40×10^{-10}	8.20×10^{-10}	9.1×10^{-11}
7.316	0.0125	1.14×10^{-8}	1.47×10^{-8}	-7.22×10^{-9}	2.45×10^{-9}	1.1×10^{-8}
7.316	0.025	1.94×10^{-6}	-6.79×10^{-8}	2.05×10^{-6}	2.82×10^{-6}	1.4×10^{-6}
7.346	0.00625	3.01×10^{-11}	-6.21×10^{-11}	5.56×10^{-11}	-1.16×10^{-10}	8.6×10^{-11}
7.376	0.00625	-7.35×10^{-10}	-7.21×10^{-10}	-7.65×10^{-10}	-6.46×10^{-10}	6.0×10^{-11}
7.376	0.0125	-3.92×10^{-9}	8.43×10^{-9}	-1.61×10^{-8}	-1.40×10^{-8}	1.2×10^{-8}
7.376	0.025	3.05×10^{-6}	1.58×10^{-6}	8.27×10^{-7}	2.72×10^{-6}	1.1×10^{-6}

$m - 2$ on radial grids shifted by δh_s ($s_{\min} \rightarrow -120 + \delta h_s$).

Origin truncation and the Newton floor. Supplementary Table 3 gives the converged Newton residual and $m - 2$ for origin truncations $s_{\text{start}} = -15$ to -30 at λ_5 and λ_6 . The offset relative to $s_{\text{start}} = -20$ decays roughly as $e^{s_{\text{start}}}$ down to -19 (the neglected local corrections are $O(e^{s_{\text{start}}})$); deeper starts raise the residual floor and add a round-off offset that is independent of λ at -24 (1.5×10^{-9} at both λ_5 and λ_7) and reaches 2.1×10^{-8} at -30 .

Angular resolution at the sixth profile. With $s_{\text{start}} = -16$, $N_b = 32 \rightarrow 48$ changes m by 2.0×10^{-8} and $N_b = 48 \rightarrow 64$ by 9×10^{-11} (`ipm_nbcheck*_lam6.out`); with $s_{\text{start}} = -20$ the $N_b = 32 \rightarrow 48$ change is $\sim 10^{-9}$ for $\lambda_4\text{--}\lambda_6$. At the seventh profile $N_b = 48 \rightarrow 64$ changes m by 2.3×10^{-9} (Extended Data Table 1 of the main text): the angular error stops converging at the floor. Tightening the Newton tolerance from 10^{-11} to 10^{-13} changes m by 1.7×10^{-11} (`ipm_noise_lam6_hs0125.out`), so the scatter at the floor is discretization, not the nonlinear solver.

Supplementary Table 3. Origin truncation and Newton residual floor.

profile	s_{start}	Newton residual $ R $	$m - 2$
λ_5	-16	1.1×10^{-13}	-5.12×10^{-9}
λ_5	-20	5.2×10^{-13}	2.32×10^{-11}
λ_5	-24	7.9×10^{-13}	-1.48×10^{-9}
λ_6	-15	7.3×10^{-13}	2.16×10^{-8}
λ_6	-16	3.6×10^{-14}	-4.92×10^{-9}
λ_6	-17	1.5×10^{-13}	-8.95×10^{-10}
λ_6	-18	2.0×10^{-13}	1.43×10^{-9}
λ_6	-19	4.8×10^{-13}	7.92×10^{-10}
λ_6	-20	4.4×10^{-13}	5.76×10^{-10}
λ_6	-26	1.9×10^{-12}	8.28×10^{-9}
λ_6	-30	1.5×10^{-11}	2.19×10^{-8}

$h_s = 0.0125$, $N_b = 32$; $\lambda_6 = 0.15092$, $\lambda_5 = 0.1706180880$.

Two independent solvers (IPM). Supplementary Table 4 compares the independent global solver with the marching solver. Converted with the crossing slopes, the differences are $0.2\text{--}3.5 \times 10^{-9}$ in m at every profile, while the difference in profile position grows by five orders of magnitude.

Supplementary Table 4. IPM profiles from two independent solvers (ipm_glob_*.log). At $h_s = 0.00625$ the global solver did not converge for $n = 5, 6$ within 15 Newton iterations (logged, not used).

n	λ_n (marching)	$ \Delta\lambda $	$ \Delta z $	$ dm/d\lambda $	$ \Delta m $
1	0.4721297348	1.4×10^{-8}	6.4×10^{-8}	2.4×10^{-1}	3.5×10^{-9}
2	0.3149618108	3.7×10^{-8}	3.8×10^{-7}	6.0×10^{-2}	2.2×10^{-9}
3	0.2415663353	3.4×10^{-8}	5.9×10^{-7}	1.1×10^{-2}	3.8×10^{-10}
4	0.1987224523	6.5×10^{-7}	1.7×10^{-5}	1.7×10^{-3}	1.1×10^{-9}
5	0.1706180880	9.1×10^{-7}	3.1×10^{-5}	2.3×10^{-4}	2.1×10^{-10}
6	0.1509200000	1.3×10^{-4}	5.9×10^{-3}	2.0×10^{-5}	2.7×10^{-9}

Independent global solver ($h_s = 0.0125$, $s_{\text{min}} = -16$, extrapolated in s_{max} ; $n = 1$ at $h_s = 0.025$, $s_{\text{min}} = -12$; $n = 4$ at $h_s = 0.00625$) against the marching solver. $|\Delta m| = |dm/d\lambda| |\Delta\lambda|$.

Supplementary Note 3. The pre-registration record

All predictions for IPM were committed to the public repository before the computation they concern, in `programs/P03_boussinesq_ladder/PREDICTIONS_IPM.md`; entries are never edited, and outcomes, deviations and corrections are appended. The time stamps are those of the commits that touch the file (`git log` on the file); a copy of the record as of submission is provided as Supplementary Data 1. Commit dates are set by the committing machine. Supplementary Data 2 therefore lists, for every commit of the repository up to the preparation of this submission, the earliest push recorded by GitHub after which the commit was on the server, in GitHub’s server time (from its public events interface). This is an upper bound on the time at which the commit became public. Every registration was on GitHub within 25 min of its commit date (Supplementary Table 5).

Deviations and corrections recorded in the file. (i) The committed λ_7 procedure needed matched scans at $h_s = 0.0125$ and 0.00625 ; the 0.0125 scan had been paused to measure the locking error, so the procedure could not complete; the converged-grid determination with four grid shifts was reported instead, judged by the same thresholds. (ii) A first phase reading of λ_7 ($z_7 = 7.3456 \pm 0.003$) was corrected the same day to 7.341 ± 0.005 (dependence on the offset) and, after the tracker repair, to 7.354 ± 0.007 ; none of these is a registered prediction. (iii) The tracker failure and its repair were recorded with a clarification that the failure counter had been logged but not gated.

Stage 4. The rule: solve $\text{Re } \Phi_0(z_n) = n\pi + \delta_n$ with $\text{Re } \Phi_0$ from the repaired deep table (smoothed by a local quadratic over ± 0.5 in z , shifted by the measured grid correction $+0.027 \pm 0.010$); offset scenarios $\delta = \delta_6 = 1.974$ (central), $\delta = 2.031$ (mean of $n = 1\text{--}5$), and a continuing drift $\delta_n = 1.974 - 0.057(n - 6)$; band = scenarios widened by the grid spread and the fit error. The required precision continues the λ_7 crossing slope with the decay per half-period predicted by $\text{Im } \Phi_0$ times $(1 + \lambda)$. A future measurement counts only if $\sigma_z \leq 0.01$; it confirms the rule at n if the measured z_n lies inside the band (widened by $2\sigma_z$), and refutes it

Supplementary Table 5. Timeline of the record. Each commit is followed by the time (UTC) by which it was on GitHub (Supplementary Data 2).

Commit	Registered	Outcome (appended later)
4dc82af 30 Sep 2026, 01:02 stage 2 entries	Stage 1, before any IPM continuation: accumulation as $\lambda \rightarrow 0$, linear law, index n , lattice of growth rates of step λ_n , rigid spectral flow, localization at the front $\lambda_3, \lambda_4, \lambda_5, \lambda_6$ and the growth rates of the fourth unstable profile from the lower profiles by fixed rules	Accumulation clause and linear law failed (spacings contract by 0.92 per profile); index n ($n = 0-4$), lattice, flow and localization held Errors 6%, 2.5%, 2%, 0.1% of a spacing; growth rates to 0.2–1.2%
c922751 30 Sep, 17:14 273ae28 30 Sep, 17:14	Stage 3: z_7, z_8 by three rung-fit rules (H1, H1', H2, H3) Stage 3, H4: $z_7 = 7.352, z_8 \approx 7.91-7.92$ (low confidence) from the phase on the coarse branch	z_7 : not decisive (main text); z_8 : untested z_7 : not decisive; the z_8 value was made with the faulty tracker (Supplementary Note 5)
f0b3f06 1 Oct, 16:19	Decision rule for λ_7 , fixed before the crossing was known	Could not run as written (scan paused); converged-grid analysis reported with the same thresholds
251bf1e 1 Oct, 19:28	Stage 4: $\lambda_8-\lambda_{10}$ from the repaired phase, with bands, required precision and a decision rule	Open benchmark (Supplementary Table 6)

otherwise. All values of $m - 2$ computed in this range before registration were at or below the floor and are disclosed in the record.

Supplementary Table 6. Stage-4 predictions and the precision a test needs (ipm_stage4_predictions.out).

n	z_n	band	λ_n	λ band	e-folds/profile	$ dm/dz $	σ_m ($\sigma_z = 0.01$)	σ_m ($\sigma_z = 0.003$)
8	7.9994	[7.9700, 8.0172]	0.125010	[0.124732, 0.125471]	2.74	1.8×10^{-9}	1.8×10^{-11}	5.3×10^{-12}
9	8.6282	[8.5895, 8.6450]	0.115899	[0.115673, 0.116421]	2.77	1.2×10^{-10}	1.2×10^{-12}	3.6×10^{-13}
10	9.2013	[9.1575, 9.2142]	0.108680	[0.108528, 0.109200]	3.06	5.9×10^{-12}	6.0×10^{-14}	1.8×10^{-14}

Supplementary Note 4. The resonance phase

Why a defect of this form has zeros [P].

Lemma. *Let $F(z) = R(z) [\cos \Theta(z) + \eta(z)]$ on $[z_0, \infty)$ with $R > 0$ and Θ continuous, Θ strictly increasing to ∞ , and $\sup |\eta| < 1$. Then F changes sign between $\Theta^{-1}(k\pi)$ and $\Theta^{-1}((k+1)\pi)$ for every admissible integer k , so it has infinitely many zeros accumulating only at $z = \infty$; if $\eta \rightarrow 0$, each zero z^* satisfies $\Theta(z^*) = (k + \frac{1}{2})\pi + O(|\eta(z^*)|)$.*

Proof. $F(\Theta^{-1}(k\pi)) = R[(-1)^k + \eta]$ has the sign of $(-1)^k$ because $|\eta| < 1$; apply the intermediate value theorem. The second statement follows from $\cos \Theta = -\eta$. $\square$

The stalled layer and the local problem [F]. As $\varepsilon \rightarrow 0$ the wall speed $D = V_1/x$ is $O(\varepsilon)$ between the stagnation point and a front. In this layer we write a perturbation of the density as $\tilde{r}(Y) e^{i\varphi}$ with $\varphi = D_0^{-1} \int \kappa ds$, $s = \ln x$, and Y the wall-normal coordinate scaled with the layer thickness, and keep the leading order in ε . For IPM this gives the local eigenproblem of the Methods,

$$[i\kappa(\hat{D} + \hat{c}Y) + \mu Y \partial_Y] \tilde{r} = -G \partial_Y \tilde{\psi} + i\kappa R_y \tilde{\psi}, \quad -\partial_Y^2 \tilde{\psi} + \kappa^2 \tilde{\psi} = -i\kappa \tilde{r},$$

with $\tilde{\psi}(0) = 0$, no growing mode, $\hat{D} = D/D_0$, $\hat{c} = G = \partial_x R$ at the wall (because $\Omega = -\partial_1 R$), μ the wall-normal strain and R_y the normal density gradient. The continued root $\kappa(s)$ defines the phase $\Phi_0 = D_0^{-1} \int \kappa ds$, integrated from the stagnation point to the front. Matching the layer solution to the stagnation-point region and to the front writes the smoothness defect as $m - 2 \simeq |A| e^{-|\text{Im } \Phi_0|} [\cos(\text{Re } \Phi_0 + \arg A) + \eta]$ with $\eta = O(\varepsilon)$, so by the Lemma the smooth profiles satisfy $\text{Re } \Phi_0(\lambda_n) = n\pi + \delta$ (main-text equation (2)), and the envelope of the defect decays by $e^{-|\Delta \text{Im } \Phi_0|}$ per half-period. The offset δ collects the $O(1)$ connection phases at the stagnation point and at the front and is fixed by the cut-off convention $D/D_0 = 2$. The derivation is formal; its hypotheses are checked numerically below.

Exact scaling in IPM [P]. Substitute $Y = \hat{D} \eta$, $\kappa = K/\hat{D}$ and $\tilde{r} = \hat{r}/\hat{D}$. Then $i\kappa(\hat{D} + \hat{c}Y) = iK(1 + \hat{c}\eta)$, $\mu Y \partial_Y = \mu \eta \partial_\eta$ and $\partial_Y = \hat{D}^{-1} \partial_\eta$, and both equations become independent of $\hat{D}$:

$$[iK(1 + \hat{c}\eta) + \mu \eta \partial_\eta] \hat{r} = -G \partial_\eta \tilde{\psi} + iK R_y \tilde{\psi}, \quad -\partial_\eta^2 \tilde{\psi} + K^2 \tilde{\psi} = -iK \hat{r}.$$

Hence $\kappa = K(G, \mu, R_y)/\hat{D}$ exactly and $\Phi_0 = \int K ds/D$. Numerically K is unchanged to 10^{-7} when $\hat{D}$ is varied from 0.012 to 2 at fixed (G, μ, R_y) (main-text Fig. 5d; `ipm_K_checks.out`). In 2D Boussinesq the vorticity obeys its own transport equation, which breaks this scaling.

The phase along the branch [N]. Supplementary Table 9 lists the repaired phase on every computed state. The offsets δ_n for four cut-offs are in Supplementary Table 7: the registered cut-off $D/D_0 = 2$ is the only one that is flat over $n = 1-5$, and every cut-off shows an extra drop of 0.04 ± 0.01 at $n = 6$, which therefore originates inside the layer. Supplementary Table 8 compares the decay of the defect per half-period with the prediction from $\text{Im } \Phi_0$.

Supplementary Table 7. Offsets δ_n for four front cut-offs (`ipm_cutoff_drift.out`).

n	cut-off 1.5	cut-off 2	cut-off 3	cut-off 5
1	1.7506	2.0331	2.3488	2.6849
2	1.7927	2.0228	2.2863	2.5643
3	1.8325	2.0320	2.2660	2.5144
4	1.8604	2.0360	2.2476	2.4712
5	1.8731	2.0301	2.2200	2.4232
6	1.8500	1.9742	2.1562	2.3253

Offsets $\delta_n = \text{Re } \Phi_0(\lambda_n) - n\pi$ with the repaired tracker for four front cut-offs D/D_0 ; no untracked points. Cut-off 2 is the registered convention.

Supplementary Table 8. Decay of the defect per half-period: prediction from $\text{Im } \Phi_0$ and measurement.

z (midpoint)	λ	predicted	measured	(meas. - pred.)/(pred. λ)
2.96	0.338	1.73	2.30	0.97
3.96	0.253	1.85	2.32	1.02
4.88	0.205	1.95	2.37	1.04
5.68	0.176	2.05	2.56	1.42
6.45	0.155	2.14	2.38	0.70

E-folds of the defect per half-period: predicted $-\pi d \text{Im } \Phi_0 / d \text{Re } \Phi_0$ from the repaired phase; measured from consecutive extrema of $|m - 2|$.

Supplementary Note 5. The tracker failure and its repair

Original tracker. Along the wall, the root κ at the next point was sought from the previous κ (and from the Hou–Luo closed form as a fallback); a point without an accepted root kept the previous κ and was counted as failed. On the shallow states no point failed.

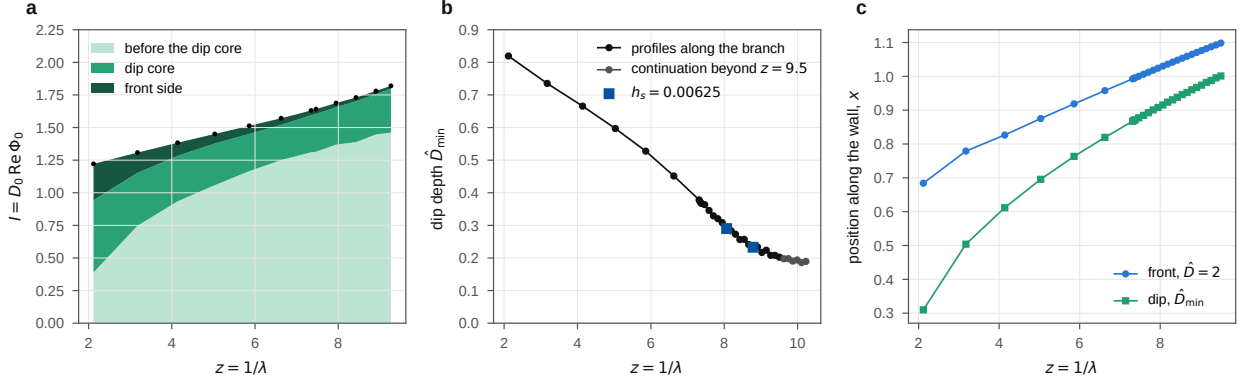
Failure. On the deep states the front side of the dip carries $\hat{D}$ from about 0.3 to 2 within a few steps of $\Delta s = 0.025$. The previous κ is then a poor predictor, the secant iteration lands on another root or fails, and holding κ inflates $K = \kappa \hat{D}$ in proportion to $\hat{D}$. Supplementary Table 10 compares $I = D_0 \text{Re } \Phi_0$ from the original and the repaired tracker; the original values are non-monotone in z and exceed the repaired ones by up to 14%.

Repair. Up to the dip the original tracking is kept. Beyond it: (i) step $\Delta s/4$; (ii) predictor $K_{\text{prev}}/\hat{D}$ (exact by the scaling of Supplementary Note 4), then the previous κ , then the Hou–Luo form; (iii) accept a root only if $|K - K_{\text{prev}}| \leq 0.25|K_{\text{prev}}|$; (iv) at a failed point hold K , not κ , and count it; (v) exclude any state with a failed point from every fit. The repaired tracker reproduces the shallow values ($\text{Re } \Phi_0 = 11.4568$ vs 11.4566 at $n = 3$; 17.7381 vs 17.7363 at $n = 5$) and fails on one of 17 deep states ($z = 9.386$, 11 points; excluded).

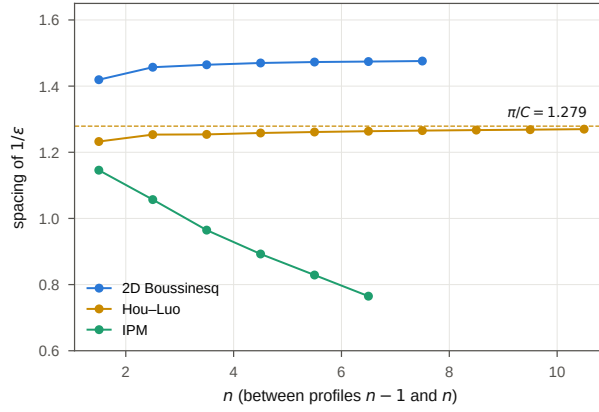
Supplementary Note 6. The deep branch and the endpoint analysis

Geometry [N]. Supplementary Table 11 and Supplementary Fig. 2 give the dip depth, the dip and front positions, the travel time $T = \int ds/\hat{D}$, the steepest wall gradient and the resolution indicator along the branch.

Supplementary Table 12 compares two deep states on $h_s = 0.0125$ and 0.00625 : the geometry and T agree to 0.3%, while the indicator, the steepest gradient and $m - 2$ change as described in the main text.



Supplementary Fig. 2. Geometry of the IPM branch. **a**, Partition of $I = D_0 \text{Re } \Phi_0$: the part before the dip core (inner layer and approach) grows as the dip and front recede; the dip core and the front side do not. **b**, Dip depth $\hat{D}_{\min}$ along the branch, with two checks at $h_s = 0.00625$. **c**, Positions of the dip and of the front ($\hat{D} = 2$).



Supplementary Fig. 3. Spacings of consecutive profiles in $1/\varepsilon$. A stalled layer of fixed extent gives increasing spacings (2D Boussinesq, Hou-Luo; dashed: the Hou-Luo asymptote $\pi/C = 1.279$ obtained from the $\lambda \rightarrow 1$ limit problem); the receding layer of IPM gives contracting spacings.

Endpoint classes [N]. The phase coefficient $I(z)$ was fitted on $z \geq 4$ with three forms: (A) $a + bz + cz^2$ (accumulation at $\lambda = 0$); (B) $a + bz + C/(z_c - z)$ and (C) $a + bz - C \ln(1 - z/z_c)$ (accumulation at $\lambda_c = 1/z_c$). Weighting the shallow points by their grid error and the deep points by their scatter, (B) and (C) prefer $z_c \approx 10\text{--}13$ with $\chi^2 \approx 33\text{--}36$ for 20 degrees of freedom, against $\chi^2 = 140$ for (A); the fits are therefore not adequate models and the preference is not significant. On the deep range alone ($z \geq 7.45$), z_c is unconstrained ($\Delta\chi^2 \leq 4$ for all $z_c \geq 9.5$), the slope of I is 0.098 and its curvature $+0.005 \pm 0.002$. Termination indicators: $\hat{D}_{\min}$ and $1/\max \partial_x R$ decrease at -0.107 and -0.050 per unit z on $z \in [7.45, 8.5]$ and at -0.023 and -0.014 over the last unit, extrapolating linearly to zero only near $z \approx 18$ and 13 (`ipm_endpoint_geometry.out`). The endpoint class is unresolved.

Supplementary Note 7. Formulations, solvers and reproducibility

Self-similar equations. With $y = x/(1 - t)^{1+\lambda}$ and $V = (1 + \lambda)y + U$, $U = \nabla^\perp \Psi$, $-\Delta \Psi = \Omega$:

- IPM: $V \cdot \nabla R = \lambda R$, $\Omega = -\partial_1 R$ ($\varepsilon = \lambda$);
- 2D Boussinesq: $\Omega + V \cdot \nabla \Omega = \partial_1 \Theta$, $V \cdot \nabla \Theta = (\lambda - 1)\Theta$ ($\varepsilon = \lambda - 1$);
- Hou-Luo: the boundary restriction of Boussinesq, with $U' = H\Omega$ (H the Hilbert transform).

All are posed on the half-plane with no penetration; by symmetry the 2D problems are solved on the quarter plane in log-polar coordinates (s, β) , $\beta = 0$ the wall.

Solver parameters. Marching solver: $N_b = 32$ Chebyshev points in β ; $h_s = 0.025, 0.0125$ (production) or 0.00625 ; $s \in [-120, 100]$; origin truncation $s_{\text{start}} = -20$; implicit Gauss–Legendre march for $s < 12$, RK4 beyond; eighth-order differences for the velocity; Newton–Krylov with central-difference products (perturbation 10^{-6} on the largest entry); tolerances 7×10^{-13} ($h_s = 0.0125$) and 1.5×10^{-12} (0.00625). Global solver: $N_b = 24$; sixth-order differences; $s_{\text{min}} = -16$ (-12 for λ_1); $s_{\text{max}} = 8, 10, 12$ with geometric extrapolation; exact sparse Jacobian; tolerance 10^{-10} .

Software and hardware. Python 3.11; NumPy 2.4; SciPy 1.17; Numba 0.67; Matplotlib 3.11. All computations ran on a four-core x86-64 machine without GPU. Logged solver time for the deep IPM campaign was about 13 core-hours; phase and geometry evaluations about one hour.

Map from results to code and data. Every figure is produced by `code/figures/make_figures.py` and every Supplementary Table by `code/analysis/make_si_tables.py`, both reading only `data/outputs`. The file `code/README.md` lists, in order, the commands that regenerate every output from scratch, with run times.

References

- [1] Chen, J. & Hou, T. Y. Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data I: analysis. Preprint at <https://arxiv.org/abs/2210.07191> (2022).
- [2] Choi, K. et al. On the finite-time blowup of a one-dimensional model for the three-dimensional axisymmetric Euler equations. *Commun. Pure Appl. Math.* **70**, 2218–2243 (2017).

Supplementary Table 9. Repaired resonance phase on every computed state
(ipm_phase3_{rungs,17,deep}.out).

z	D_0	$\hat{D}_{\text{dip}}$	s_{cut}	$\text{Re } \Phi_0$	$\text{Im } \Phi_0$	I	untracked
2.1181	0.23606	0.8194	-0.380	5.1747	-1.9087	1.22156	0
3.1750	0.15748	0.7353	-0.250	8.3059	-3.5978	1.30803	0
4.1396	0.12078	0.6657	-0.191	11.4568	-5.4154	1.38378	0
5.0321	0.09936	0.5971	-0.133	14.6024	-7.3379	1.45091	0
5.8610	0.08531	0.5276	-0.085	17.7381	-9.3555	1.51322	0
6.6260	0.07546	0.4515	-0.043	20.8238	-11.4252	1.57136	0
7.3160	0.06834	0.3750	-0.007	23.8203	-13.5550	1.62796	0
7.3460	0.06806	0.3717	-0.006	23.9580	-13.6584	1.63068	0
7.3760	0.06779	0.3674	-0.004	24.0928	-13.7557	1.63319	0
7.4660	0.06697	0.3636	+0.000	24.5106	-14.0662	1.64148	0
7.5860	0.06591	0.3458	+0.006	25.0602	-14.4663	1.65174	0
7.7060	0.06488	0.3293	+0.012	25.6446	-14.9372	1.66393	0
7.8260	0.06389	0.3204	+0.018	26.2192	-15.3738	1.67514	0
7.9460	0.06292	0.3083	+0.024	26.8226	-15.8768	1.68781	0
8.0660	0.06199	0.2902	+0.030	27.4185	-16.3487	1.69964	0
8.1860	0.06108	0.2834	+0.035	27.9493	-16.7308	1.70714	0
8.3060	0.06020	0.2729	+0.041	28.6510	-17.3721	1.72472	0
8.4260	0.05934	0.2565	+0.047	29.1547	-17.6826	1.73005	0
8.5460	0.05851	0.2570	+0.052	29.7744	-18.1999	1.74201	0
8.6660	0.05770	0.2413	+0.058	30.4093	-18.7190	1.75452	0
8.7860	0.05691	0.2322	+0.063	31.0771	-19.3427	1.76856	0
8.9060	0.05614	0.2339	+0.068	31.7197	-19.8606	1.78080	0
9.0260	0.05540	0.2170	+0.073	32.3954	-20.5066	1.79456	0
9.1460	0.05467	0.2238	+0.079	33.0543	-21.0476	1.80703	0
9.2660	0.05396	0.2080	+0.084	33.7243	-21.6692	1.81979	0
9.3860	0.05327	0.2081	+0.089	34.6567	-22.5874	1.84619	11 (excluded)

Rungs $n = 1-5$ and λ_6 on $h_s = 0.0125$; $z = 7.316-7.376$ on $h_s = 0.00625$; deep states on $h_s = 0.0125$ (continuation).
 $I = D_0 \text{Re } \Phi_0$.

Supplementary Table 10. Phase coefficient from the original and the repaired tracker
(ipm_wkb_e5_cut2.out, ipm_tracker_diag.out).

z	I , original tracker	failed points (counted)	I , repaired tracker
6.626	1.57134	0	1.57136
6.746	1.58478	0	—
6.866	1.59341	0	—
7.466	1.64395	0	1.64148
7.586	1.65870	2	1.65174
7.706	1.72326	3	1.66393
7.826	1.74614	3	1.67514
7.946	1.92056	4	1.68781
8.066	1.74702	3	1.69964
8.186	1.85564	4	1.70714
8.306	1.88404	4	1.72472
8.426	1.91985	4	1.73005
8.546	1.95728	4	1.74201
8.666	1.98989	4	1.75452

Supplementary Table 11. Geometry along the IPM branch (ipm_travel_time.out, ipm_deep_e5.out).

z	$\hat{D}_{\min}$	x_{dip}	x_{cut}	T	$\max \partial_x R$	indicator (%)
2.118	0.8194	0.3098	0.6841	9.6691	—	—
3.175	0.7353	0.5040	0.7790	9.8360	—	—
4.140	0.6657	0.6117	0.8265	9.9403	—	—
5.032	0.5971	0.6956	0.8753	10.0315	—	—
5.861	0.5276	0.7633	0.9186	10.1126	—	—
6.626	0.4515	0.8195	0.9577	10.1880	—	—
7.316	0.3781	0.8673	0.9926	10.2602	—	—
7.346	0.3717	0.8696	0.9941	10.2635	—	—
7.376	0.3674	0.8715	0.9956	10.2668	—	—
7.466	0.3636	0.8774	1.0001	10.2768	7.83	3.3
7.586	0.3458	0.8846	1.0061	10.2903	8.14	3.4
7.706	0.3293	0.8925	1.0121	10.3042	8.77	3.7
7.826	0.3204	0.9007	1.0181	10.3183	9.32	4.7
7.946	0.3083	0.9080	1.0241	10.3326	9.66	4.6
8.066	0.2902	0.9154	1.0301	10.3471	10.46	5.1
8.186	0.2834	0.9236	1.0360	10.3619	11.11	6.3
8.306	0.2729	0.9305	1.0419	10.3767	11.47	5.4
8.426	0.2565	0.9380	1.0477	10.3917	12.50	6.9
8.546	0.2570	0.9461	1.0535	10.4067	13.08	7.8
8.666	0.2413	0.9523	1.0593	10.4217	13.74	5.9
8.786	0.2322	0.9601	1.0649	10.4366	14.72	9.1
8.906	0.2339	0.9672	1.0706	10.4514	14.88	7.8
9.026	0.2170	0.9739	1.0762	10.4661	16.22	9.1
9.146	0.2238	0.9820	1.0818	10.4806	16.68	10.0
9.266	0.2080	0.9875	1.0874	10.4949	17.61	8.1
9.386	0.2081	0.9955	1.0928	10.5091	18.35	11.0
9.506	0.2023	1.0010	1.0983	10.5229	18.93	6.9
9.626	0.1975	—	—	—	19.93	12.0
9.746	0.1980	—	—	—	20.26	7.3
9.866	0.1905	—	—	—	21.42	13.0
9.986	0.1939	—	—	—	21.62	8.0
10.106	0.1858	—	—	—	22.83	13.0
10.226	0.1893	—	—	—	23.06	7.9

$T = \int_{-10}^{s_{\text{cut}}} ds / \hat{D}$. Indicator: $\max |\Omega_b + \partial_x R| / \max \partial_x R$ over the dip-to-front window (deep continuation, $h_s = 0.0125$).

Supplementary Table 12. Resolution checks of deep states (ipm_deepres_z*.out).

z	h_s	$m - 2$	$\hat{D}_{\min}$	$\max \partial_x R$	indicator (%)	T	$\text{Re } \Phi_0$
8.0660	0.0125	2.45×10^{-8}	0.2902	10.46	5.1	10.3471	27.4186
8.0660	0.00625	1.04×10^{-9}	0.2902	10.85	1.4	10.3472	27.4556
8.7860	0.0125	3.25×10^{-7}	0.2322	14.72	9.1	10.4366	31.0771
8.7860	0.00625	2.01×10^{-9}	0.2329	15.63	2.5	10.4366	31.0941